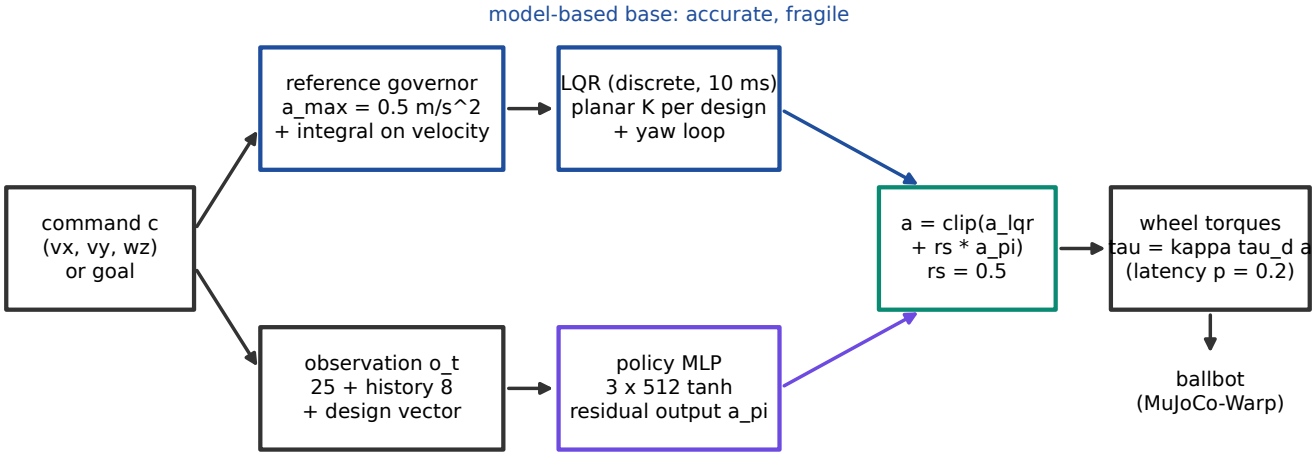
model-based base: accurate, fragile
reference governor
a_max = 0.5 m/s^2
+ integral on velocity
LQR (discrete, 10 ms)
planar K per design
+ yaw loop
command c
(vx, vy, wz)
or goal
a = clip(a_lqr
+ rs * a_pi)
rs = 0.5
wheel torques
tau = kappa tau_d a
(latency p = 0.2)
ballbot
(MuJoCo-Warp)
observation o_t
25 + history 8
+ design vector
policy MLP
3 x 512 tanh
residual output a_pi
learned residual: bounded, trained with PPO